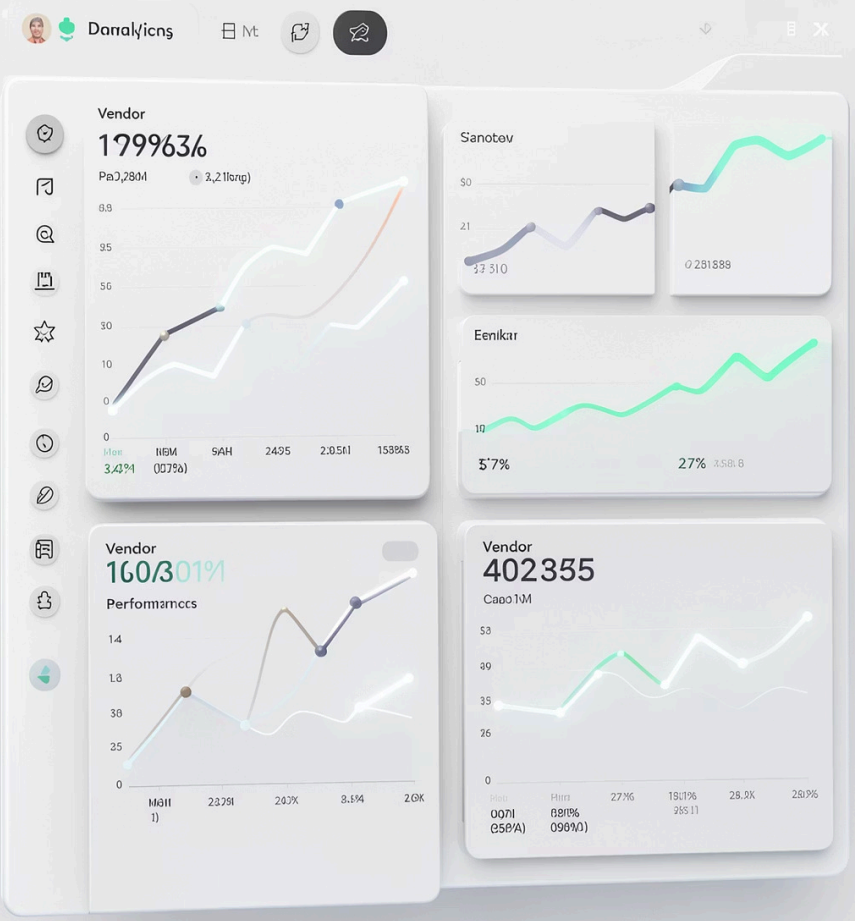
# Vendor Performance Analysis

Exploratory Data Analysis · Key Findings · Strategic Recommendations

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# Red Flags in the Data

## Gross Profit

Min: **-$52,002** — losses from heavy discounts or below-cost pricing

## Zero Sales

Products purchased but never sold — slow-moving or obsolete stock

## Stock Turnover

Range: **0 – 274.5** — some items never move, others sell fast

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# Outliers Detected

### Freight Cost

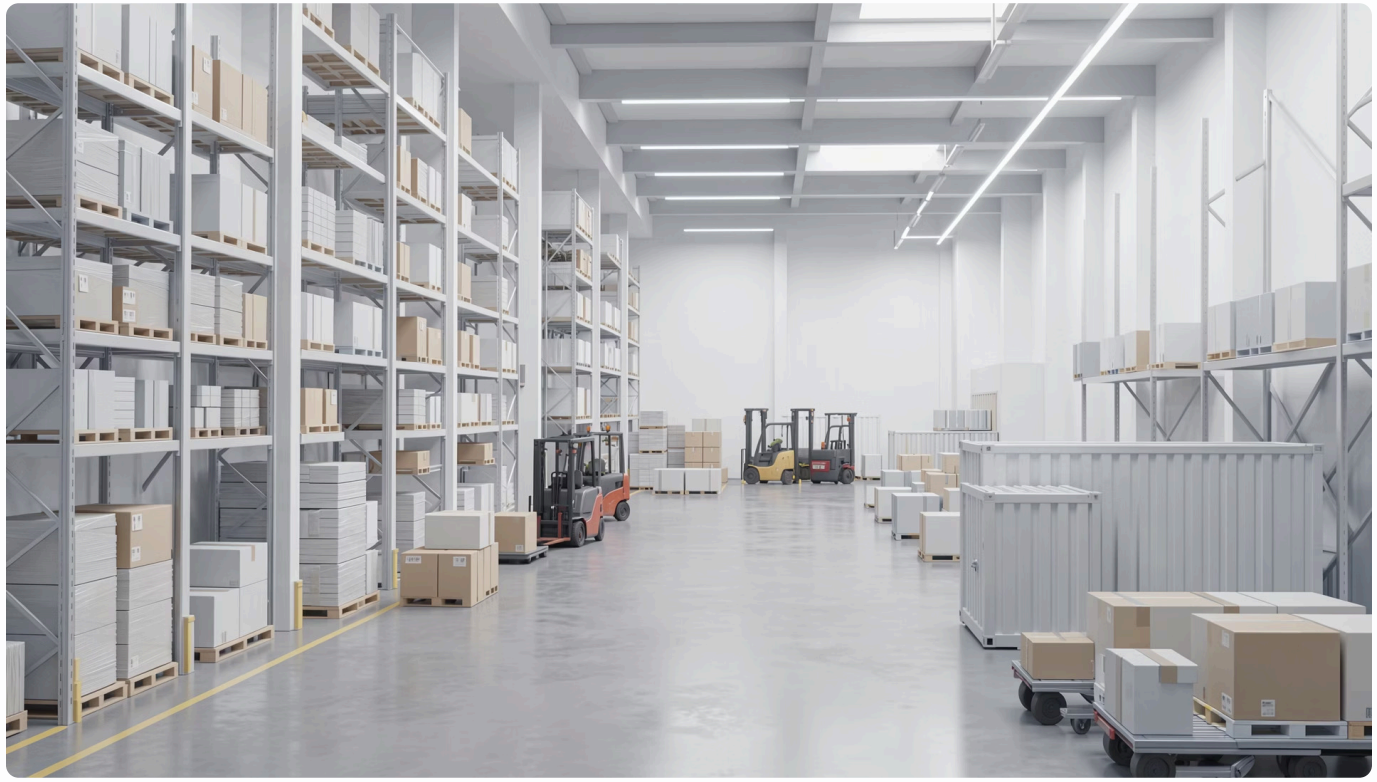
Range: $0.09 – $257,032 — logistics inefficiencies or bulk shipments

### Purchase Prices

Max $5,681 vs. mean $24 — premium product outliers

### Actual Prices

Max $7,499 vs. mean $35 — significant price variation



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# Data Filtering Applied

To ensure reliable insights, we excluded inconsistent data points:

→ Gross Profit < 0

Remove loss-making transactions

→ Profit Margin < 0

Focus on profitable transactions only

→ Total Sales Quantity = 0

Eliminate unsold inventory

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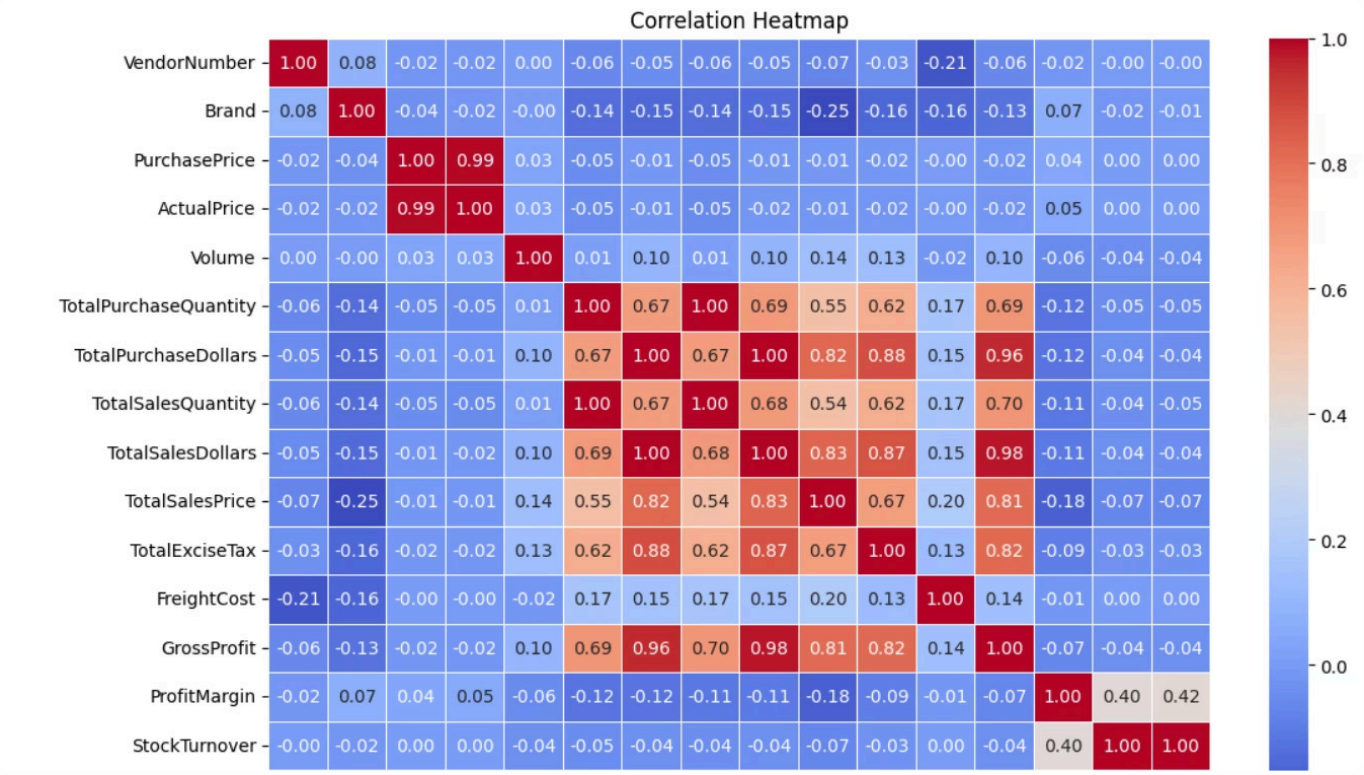
# Correlation Insights




### Purchase Price → Revenue

Weak: **-0.012**

### Purchase Qty → Sales Qty

Strong: **0.999** — efficient turnover

### Turnover → Profitability

Weak: **-0.038** — speed ≠ profit

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Brands with Low sales but High Profit Margins:
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| | Description | TotalSalesDollars | ProfitMargin |
|---|---|---|---|
| **6199** | Santa Rita Organic Svgn Bl | 9.99 | 66.466466 |
| **2369** | Debauchery Pnt Nr | 11.58 | 65.975820 |
| **2070** | Concannon Glen Ellen Wh Zin | 15.95 | 83.448276 |
| **2188** | Crown Royal Apple | 27.86 | 89.806174 |
| **6237** | Sauza Sprklg Wild Berry Marg | 27.96 | 82.153076 |
| **...** | ... | ... | ... |
| **5074** | Nanbu Bijin Southern Beauty | 535.68 | 76.747312 |
| **2271** | Dad's Hat Rye Whiskey | 538.89 | 81.851584 |
| **57** | A Bichot Clos Marechaudes | 539.94 | 67.740860 |
| **6245** | Sbragia Home Ranch Merlot | 549.75 | 66.444748 |
| **3326** | Goulee Cos d'Estournel 10 | 558.87 | 69.434752 |

# 198 Brands: Low Sales, High Margins

These brands have strong profitability but underperform on volume — prime candidates for targeted promotions and pricing optimization.

**Opportunity:** Increase sales volume without sacrificing margin through marketing and price adjustments.

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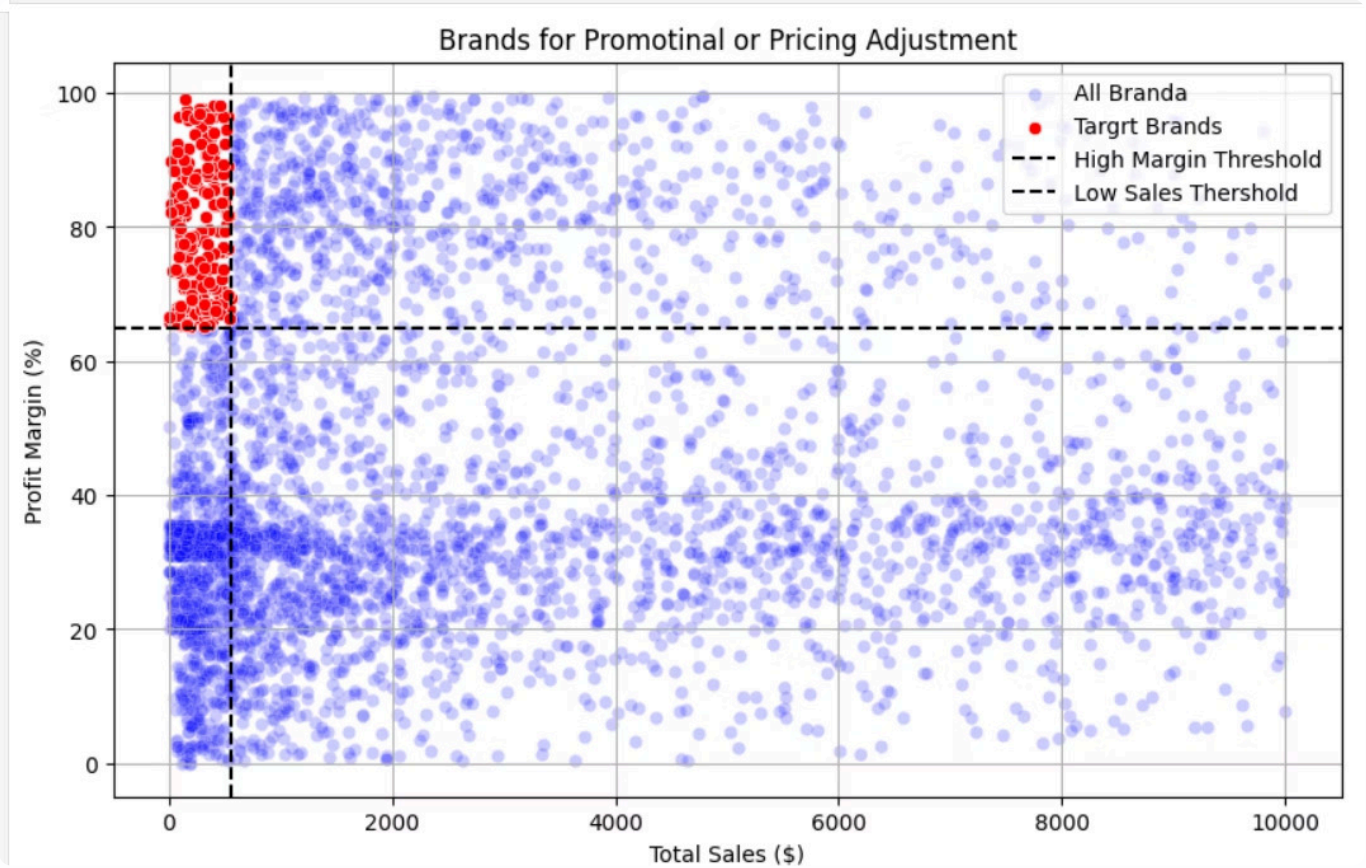
# Vendor Concentration Risk




## Top 10 Vendors

Contribute **65.69%** of total purchases

## Remaining Vendors

Contribute only **34.32%**

⚠ Over-reliance on few vendors introduces supply chain risk — diversification needed.

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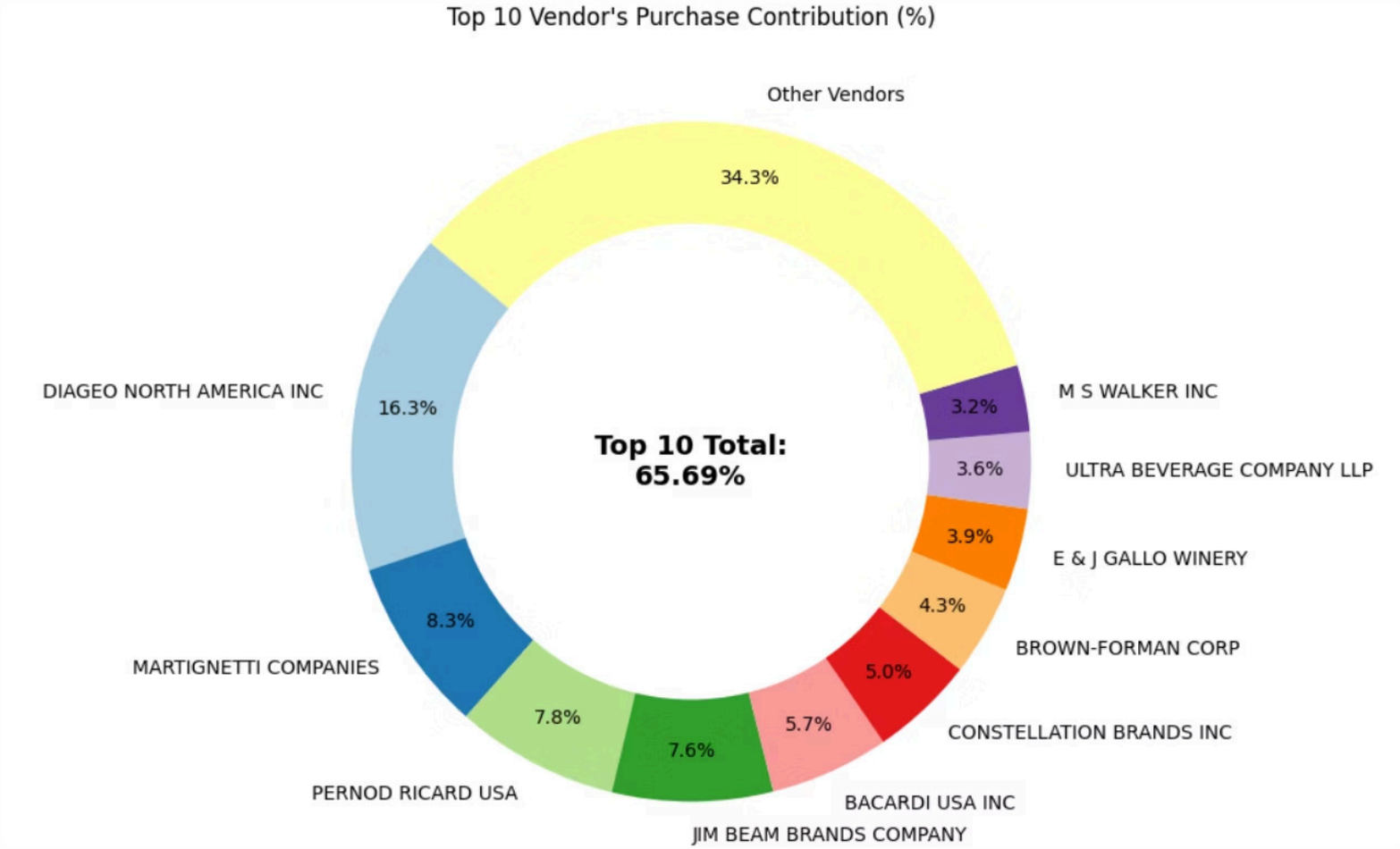


# Bulk Purchasing & Inventory Risk

## Bulk Discount

Large orders receive **72% lower** unit cost — $10.78/unit vs. smaller orders

## Unsold Inventory

**$2.71M** tied up in slow-moving stock — increases storage costs and reduces cash flow

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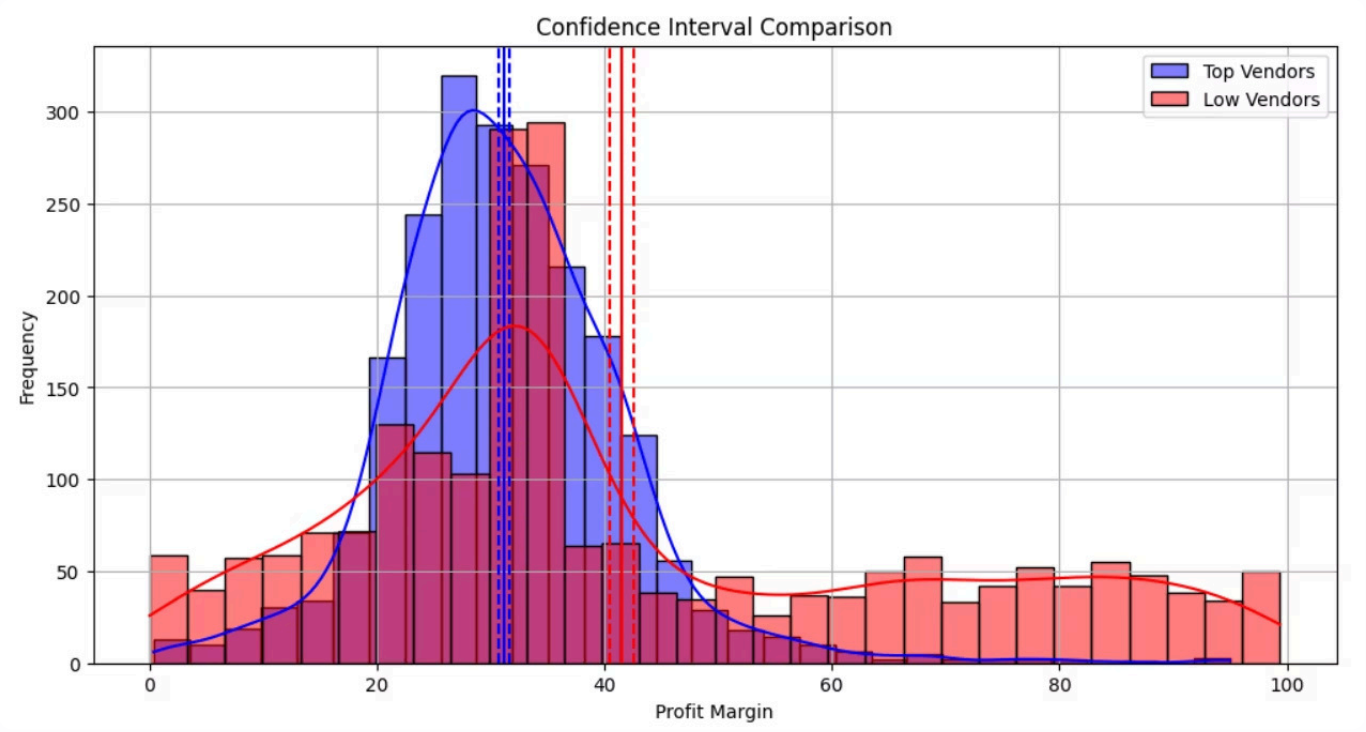
# Profit Margin: High vs. Low Performers




## Top Vendors

Mean: **31.17%**

95% CI: 30.74% – 31.61%

## Low Vendors

Mean: **41.55%**

95% CI: 40.48% – 42.62%

Hypothesis test: **H0 rejected** — two groups operate under distinctly different profitability models.

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# Final Recommendations

01

Re-price low-sales, high-margin brands

02

Diversify vendor partnerships

03

Leverage bulk purchasing

04

Clear slow-moving inventory

05

Boost marketing for low-performers

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